

Class Notes on Duality in Linear Programming

For B.Tech Students

1 Introduction

Every Linear Programming Problem (LPP) is associated with another linear programming problem called its **dual problem**.

The original problem is called the **primal problem** and the associated problem is called its **dual**.

Thus,

Primal Problem	$\longleftrightarrow$	Dual Problem
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The primal and dual represent two different ways of looking at the same optimization problem. For example, the primal may ask:

How many units of different products should be produced to maximize profit?

whereas the dual may ask:

What values should be assigned to the available resources?

Example of Primal and Dual

Consider a company producing two products, Product 1 and Product 2.

Let x_1 = number of units of Product 1, x_2 = number of units of Product 2. Suppose the profit per unit is Product 1: Rs. 3, Product 2: Rs. 2. The company uses two resources. The amount of each resource required for producing one unit of each product is given below.

	Product 1	Product 2	Available
Resource 1	1	1	4
Resource 2	2	1	6

The company wants to decide how many units of each product should be produced so that the total profit is maximum.

Primal Problem

The profit from Product 1 is Rs. 3 per unit and the profit from Product 2 is Rs. 2 per unit. Therefore, the total profit is $Z_P = 3x_1 + 2x_2$. Hence, the objective is Maximize $Z_P = 3x_1 + 2x_2$. Now consider the resource limitations. For Resource 1, one unit of Product 1 requires one unit of the resource and one unit of Product 2 also requires one unit. Therefore, the total use of Resource 1 is $x_1 + x_2$. Only 4 units of Resource 1 are available,

so $x_1 + x_2 \leq 4$. For Resource 2, one unit of Product 1 requires 2 units and one unit of Product 2 requires 1 unit. Therefore, the total use of Resource 2 is $2x_1 + x_2$. Only 6 units are available, so $2x_1 + x_2 \leq 6$. Also, negative production is not possible. Hence, $x_1, x_2 \geq 0$. Therefore, the primal problem is

$\begin{array}{ll} \text{Maximize} & Z_P = 3x_1 + 2x_2, \\ \text{subject to} & x_1 + x_2 \leq 4, \\ & 2x_1 + x_2 \leq 6, \\ & x_1, x_2 \geq 0. \end{array}$

Here,

Primal problem = how much of each product should be produced?

Formation of the Dual Problem

The primal problem has two resource constraints. Therefore, we introduce two dual variables: w_1 = value of one unit of Resource 1, and w_2 = value of one unit of Resource 2. The dual problem tries to assign values to the resources.

Since 4 units of Resource 1 are available and 6 units of Resource 2 are available, the total value of the resources is $4w_1 + 6w_2$. Therefore, the dual objective is Minimize $Z_D = 4w_1 + 6w_2$. We minimize because we want to find the smallest total value of the resources that is sufficient to account for the profits from the products.

First Dual Constraint

To produce one unit of Product 1, we require 1 unit of Resource 1 and 2 units of Resource 2. Therefore, the value of the resources used in one unit of Product 1 is $w_1 + 2w_2$. The profit from one unit of Product 1 is Rs. 3.

Hence, the resource value must be at least equal to its profit: $w_1 + 2w_2 \geq 3$.

Second Dual Constraint

To produce one unit of Product 2, we require 1 unit of Resource 1 and 1 unit of Resource 2. Therefore, the value of the resources used in one unit of Product 2 is $w_1 + w_2$. The profit from one unit of Product 2 is Rs. 2.

Hence, $w_1 + w_2 \geq 2$. Also, $w_1, w_2 \geq 0$. Therefore, the dual problem is

$\begin{array}{ll} \text{Minimize} & Z_D = 4w_1 + 6w_2, \\ \text{subject to} & w_1 + 2w_2 \geq 3, \\ & w_1 + w_2 \geq 2, \\ & w_1, w_2 \geq 0. \end{array}$

Interpretation

The primal problem asks:

How many units of Product 1 and Product 2 should be produced to obtain maximum profit?

The dual problem asks:

What values should be assigned to the available resources so that their total value is minimum while still being sufficient to account for the profit of each product?

2 Symmetric Dual Problems

We first consider the simplest and most important primal–dual pair.

Suppose the primal problem is

$$\begin{array}{ll} \text{Maximize} & Z_P = c_1x_1 + c_2x_2 + \cdots + c_nx_n, \\ \text{subject to} & a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \leq b_1, \\ & a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \leq b_2, \\ & \vdots \\ & a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \leq b_m, \\ & x_1, x_2, \dots, x_n \geq 0. \end{array}$$

Then its dual is

$$\begin{array}{ll} \text{Minimize} & Z_D = b_1w_1 + b_2w_2 + \cdots + b_mw_m, \\ \text{subject to} & a_{11}w_1 + a_{21}w_2 + \cdots + a_{m1}w_m \geq c_1, \\ & a_{12}w_1 + a_{22}w_2 + \cdots + a_{m2}w_m \geq c_2, \\ & \vdots \\ & a_{1n}w_1 + a_{2n}w_2 + \cdots + a_{mn}w_m \geq c_n, \\ & w_1, w_2, \dots, w_m \geq 0. \end{array}$$

Thus,

Max with $\leq$	$\longleftrightarrow$	Min with $\geq$
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3 Matrix Form

The primal problem

$$\begin{array}{ll} \text{Maximize} & Z_P = c^T x, \\ \text{subject to} & Ax \leq b, \\ & x \geq 0 \end{array}$$

has the dual

$$\begin{array}{ll} \text{Minimize} & Z_D = b^T w, \\ \text{subject to} & A^T w \geq c, \\ & w \geq 0. \end{array}$$

The important transformation is

$$A \longrightarrow A^T.$$

Therefore, the rows of the primal coefficient matrix become columns of the dual coefficient matrix.

Type of LPP	Standard Form of Constraints
Maximization problem	$\leq$
Minimization problem	$\geq$

$$\begin{array}{ll} \text{Maximization:} & Ax \leq b, \quad x \geq 0, \\ \text{Minimization:} & Ax \geq b, \quad x \geq 0. \end{array}$$

Example 1: Standard Minimization Problem

Consider the problem

$$\begin{array}{ll} \text{Minimize} & Z_P = 3x_1 + x_2, \\ \text{subject to} & 2x_1 + 3x_2 \geq 2, \\ & x_1 + x_2 \geq 1, \\ & x_1, x_2 \geq 0. \end{array}$$

This is already in standard minimization form.

In matrix form,

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad c = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Since the primal is a minimization problem, its dual is a maximization problem.

Introduce two dual variables w_1, w_2 .

The dual objective function is $Z_D = 2w_1 + w_2$. The transpose of A is

$$A^T = \begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix}.$$

Therefore, the dual is

$\begin{array}{ll} \text{Maximize} & Z_D = 2w_1 + w_2, \\ \text{subject to} & 2w_1 + w_2 \leq 3, \\ & 3w_1 + w_2 \leq 1, \\ & w_1, w_2 \geq 0. \end{array}$

Hence,

$\text{Min with } \geq \longleftrightarrow \text{Max with } \leq .$

4 Standard Form of the Primal

According to the standard form used here:

Maximization Problem

For a maximization problem, all constraints are written with $\boxed{\leq}$ so that

$$\begin{array}{ll} \text{Maximize} & c^T x, \\ \text{subject to} & Ax \leq b, \\ & x \geq 0. \end{array}$$

Minimization Problem

For a minimization problem, all constraints are written with $\boxed{\geq}$ so that

$$\begin{array}{ll} \text{Minimize} & c^T x, \\ \text{subject to} & Ax \geq b, \\ & x \geq 0. \end{array}$$

This is particularly useful when a problem contains mixed constraints.

5 Unsymmetrical Dual Problem

Suppose the primal problem contains equality constraints:

$$\begin{array}{ll} \text{Maximize} & Z_P = c^T x, \\ \text{subject to} & Ax = b, \\ & x \geq 0. \end{array}$$

Then its dual is

$$\begin{array}{ll} \text{Minimize} & Z_D = b^T w, \\ \text{subject to} & A^T w \geq c. \end{array}$$

In this case, the dual variables $w_1, w_2, \dots, w_m$ are **unrestricted in sign**.

Thus, Equality constraint in primal $\implies$ corresponding dual variable unrestricted.

Example 2: Primal Containing Equality Constraints

Consider

$$\begin{array}{ll} \text{Maximize} & Z_P = 2x_1 + 3x_2 + x_3, \\ \text{subject to} & 4x_1 + 3x_2 + x_3 = 6, \\ & x_1 + 2x_2 + 5x_3 = 4, \\ & x_1, x_2, x_3 \geq 0. \end{array}$$

There are two equality constraints. Therefore, introduce two dual variables w_1, w_2 , and both are unrestricted in sign.

The right-hand-side constants 6 and 4 form the dual objective function: $Z_D = 6w_1 + 4w_2$. There are three primal variables, so there will be three dual constraints.

Using the columns of the primal coefficient matrix, we obtain $4w_1 + w_2 \geq 2$, $3w_1 + 2w_2 \geq 3$, and $w_1 + 5w_2 \geq 1$. Therefore, the dual is

$$\begin{array}{ll} \text{Minimize} & Z_D = 6w_1 + 4w_2, \\ \text{subject to} & 4w_1 + w_2 \geq 2, \\ & 3w_1 + 2w_2 \geq 3, \\ & w_1 + 5w_2 \geq 1, \\ & w_1, w_2 \text{ unrestricted in sign.} \end{array}$$

This example gives the important rule Primal equality $\longleftrightarrow$ unrestricted dual variable.

6 Dual of a Mixed System

An LPP may contain $\leq$, $\geq$, $=$ constraints together with unrestricted variables.

The safest method is:

First convert the problem into standard primal form.

Then construct the dual.

6.1 Rule 1: Changing the Direction of an Inequality

If a maximization problem contains $a_1x_1 + \cdots + a_nx_n \geq b$, multiply the entire constraint by -1 : $-a_1x_1 - \cdots - a_nx_n \leq -b$. Similarly, for a minimization problem, a $\leq$ constraint may be converted into a $\geq$ constraint by multiplying by -1 . For example, $2x_1 + x_2 \leq 6$ can be written as $-2x_1 - x_2 \geq -6$.

6.2 Rule 2: Equality Constraint

An equality $a_1x_1 + \cdots + a_nx_n = b$ is equivalent to the two inequalities $a_1x_1 + \cdots + a_nx_n \leq b$ and $a_1x_1 + \cdots + a_nx_n \geq b$. Thus an equality can be replaced by two inequalities in opposite directions.

6.3 Rule 3: Unrestricted Variable

If a variable x_j is unrestricted in sign, write

$$x_j = x'_j - x''_j$$

where

$$x'_j, x''_j \geq 0.$$

This converts the unrestricted variable into two non-negative variables.

Example 3: Mixed System

Consider the problem

$$\begin{aligned} &\text{Minimize} && Z_P = 2x_2 + 5x_3, \\ &\text{subject to} && x_1 + x_2 \geq 2, \\ &&& 2x_1 + x_2 + 6x_3 \leq 6, \\ &&& x_1 - x_2 + 3x_3 = 4, \\ &&& x_1, x_2, x_3 \geq 0. \end{aligned}$$

Since this is a minimization problem, we first convert all constraints to the form $\geq$.

The first constraint is already $x_1 + x_2 \geq 2$. The second constraint $2x_1 + x_2 + 6x_3 \leq 6$ becomes $-2x_1 - x_2 - 6x_3 \geq -6$. The equality $x_1 - x_2 + 3x_3 = 4$ is replaced by $x_1 - x_2 + 3x_3 \geq 4$ and $x_1 - x_2 + 3x_3 \leq 4$. Multiplying the second of these by -1 gives $-x_1 + x_2 - 3x_3 \geq -4$. Hence the standard primal form is

$$\begin{aligned}
& \text{Minimize} && Z_P = 0x_1 + 2x_2 + 5x_3, \\
& \text{subject to} && x_1 + x_2 \geq 2, \\
& && -2x_1 - x_2 - 6x_3 \geq -6, \\
& && -x_1 + x_2 - 3x_3 \geq -4, \\
& && x_1 - x_2 + 3x_3 \geq 4, \\
& && x_1, x_2, x_3 \geq 0.
\end{aligned}$$

Introduce dual variables $w_1, w_2, w'_3, w''_3 \geq 0$. The dual is

$$\begin{aligned}
& \text{Maximize} && Z_D = 2w_1 - 6w_2 - 4w'_3 + 4w''_3, \\
& \text{subject to} && w_1 - 2w_2 - w'_3 + w''_3 \leq 0, \\
& && w_1 - w_2 + w'_3 - w''_3 \leq 2, \\
& && -6w_2 - 3w'_3 + 3w''_3 \leq 5.
\end{aligned}$$

Now put $w_3 = w'_3 - w''_3$. Since w'_3 and w''_3 are non-negative, w_3 is unrestricted in sign. Therefore, the dual can be written more simply as

$ \begin{aligned} & \text{Maximize} && Z_D = 2w_1 - 6w_2 - 4w_3, \\ & \text{subject to} && w_1 - 2w_2 - w_3 \leq 0, \\ & && w_1 - w_2 + w_3 \leq 2, \\ & && -6w_2 - 3w_3 \leq 5, \\ & && w_1, w_2 \geq 0, \\ & && w_3 \text{ unrestricted in sign.} \end{aligned} $

This example shows why an equality constraint in the primal produces an unrestricted variable in the dual.

Example 4: Unrestricted Primal Variable

Consider the linear programming problem

$$\begin{aligned}
& \text{Minimize} && Z_P = x_1 + x_2 + x_3, \\
& \text{subject to} && x_1 - 3x_2 + 4x_3 = 5, \\
& && x_1 - 2x_2 \leq 3, \\
& && 2x_2 - x_3 \geq 4, \\
& && x_1, x_2 \geq 0,
\end{aligned}$$

where x_3 is unrestricted in sign.

We shall first convert the problem into standard form and then construct its dual.

Step 1: Replace the unrestricted variable

Since x_3 is unrestricted in sign, write $x_3 = x'_3 - x''_3$, where $x'_3, x''_3 \geq 0$. Substituting this in the objective function gives $Z_P = x_1 + x_2 + x'_3 - x''_3$. The constraints become $x_1 - 3x_2 + 4x'_3 - 4x''_3 = 5$, $x_1 - 2x_2 \leq 3$, and $2x_2 - x'_3 + x''_3 \geq 4$. Thus,

$$\begin{array}{ll} \text{Minimize} & Z_P = x_1 + x_2 + x'_3 - x''_3, \\ \text{subject to} & x_1 - 3x_2 + 4x'_3 - 4x''_3 = 5, \\ & x_1 - 2x_2 \leq 3, \\ & 2x_2 - x'_3 + x''_3 \geq 4, \\ & x_1, x_2, x'_3, x''_3 \geq 0. \end{array}$$

Step 2: Convert the constraints into standard minimization form

For a minimization problem, the standard form of the constraints is $\boxed{\geq}$. The equality constraint $x_1 - 3x_2 + 4x'_3 - 4x''_3 = 5$ is equivalent to $x_1 - 3x_2 + 4x'_3 - 4x''_3 \geq 5$ and $x_1 - 3x_2 + 4x'_3 - 4x''_3 \leq 5$. Multiplying the second inequality by -1 , we obtain $-x_1 + 3x_2 - 4x'_3 + 4x''_3 \geq -5$. Also, $x_1 - 2x_2 \leq 3$ can be written as $-x_1 + 2x_2 \geq -3$. Therefore, the primal in standard minimization form is

$\begin{array}{ll} \text{Minimize} & Z_P = x_1 + x_2 + x'_3 - x''_3, \\ \text{subject to} & x_1 - 3x_2 + 4x'_3 - 4x''_3 \geq 5, \\ & -x_1 + 3x_2 - 4x'_3 + 4x''_3 \geq -5, \\ & -x_1 + 2x_2 \geq -3, \\ & 2x_2 - x'_3 + x''_3 \geq 4, \\ & x_1, x_2, x'_3, x''_3 \geq 0. \end{array}$

Step 3: Introduce the dual variables

Introduce the dual variables $y_1, y_2, y_3, y_4 \geq 0$ corresponding respectively to the four primal constraints. Since the primal is a minimization problem with $\geq$ constraints, the dual is a maximization problem with $\leq$ constraints. Thus, Min with $\geq \iff$ Max with $\leq$. The right-hand sides of the primal constraints are 5, -5 , -3 , 4. Hence, the dual objective function is Maximize $Z_D = 5y_1 - 5y_2 - 3y_3 + 4y_4$.

Step 4: Form the dual constraints

There is one dual constraint for each primal variable. **For x_1 :** The coefficients of x_1 in the four primal constraints are 1, -1 , -1 , 0. The coefficient of x_1 in the primal objective function is 1. Therefore, $y_1 - y_2 - y_3 \leq 1$. **For x_2 :**

The coefficients of x_2 are -3 , 3, 2, -2 . Hence, $-3y_1 + 3y_2 + 2y_3 + 2y_4 \leq 1$. **For x'_3 :**

The coefficients are 4, -4 , 0, -1 . Since the coefficient of x'_3 in the primal objective function is 1, we obtain $4y_1 - 4y_2 - y_4 \leq 1$.

For x_3'' : The coefficients are $-4, \quad 4, \quad 0, \quad 1$. The coefficient of x_3'' in the primal objective function is -1 . Therefore, $-4y_1 + 4y_2 + y_4 \leq -1$. Thus, the dual is

$$\begin{aligned} & \text{Maximize} && Z_D = 5y_1 - 5y_2 - 3y_3 + 4y_4, \\ & \text{subject to} && y_1 - y_2 - y_3 \leq 1, \\ & && -3y_1 + 3y_2 + 2y_3 + 2y_4 \leq 1, \\ & && 4y_1 - 4y_2 - y_4 \leq 1, \\ & && -4y_1 + 4y_2 + y_4 \leq -1, \\ & && y_1, y_2, y_3, y_4 \geq 0. \end{aligned}$$

Step 5: Simplification

Since the first primal constraint was originally an equality, define $y = y_1 - y_2$. Since $y_1, y_2 \geq 0$, the variable y is unrestricted in sign.

The dual objective function becomes $Z_D = 5y - 3y_3 + 4y_4$. The first two dual constraints become $y - y_3 \leq 1$, and $-3y + 2y_3 + 2y_4 \leq 1$. Now consider the last two constraints: $4y - y_4 \leq 1$, and $-4y + y_4 \leq -1$. Multiplying the second inequality by -1 gives $4y - y_4 \geq 1$. Hence, $4y - y_4 \leq 1$ and $4y - y_4 \geq 1$. Therefore, $\boxed{4y - y_4 = 1}$. Thus, the dual can finally be written as

$\begin{aligned} & \text{Maximize} && Z_D = 5y - 3y_3 + 4y_4, \\ & \text{subject to} && y - y_3 \leq 1, \\ & && -3y + 2y_3 + 2y_4 \leq 1, \\ & && 4y - y_4 = 1, \\ & && y \text{ unrestricted in sign,} \\ & && y_3, y_4 \geq 0. \end{aligned}$

Direct Form of the Dual

If we denote the dual variables corresponding to the original three constraints by w_1, w_2, w_3 , then the same dual may be written more compactly as

$\begin{aligned} & \text{Maximize} && Z_D = 5w_1 + 3w_2 + 4w_3, \\ & \text{subject to} && w_1 + w_2 \leq 1, \\ & && -3w_1 - 2w_2 + 2w_3 \leq 1, \\ & && 4w_1 - w_3 = 1, \\ & && w_1 \text{ unrestricted in sign,} \\ & && w_2 \leq 0, \\ & && w_3 \geq 0. \end{aligned}$
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Here, w_1 is unrestricted because the first primal constraint is an equality, $w_2 \leq 0$ because the second primal constraint has the sign $\leq$ in a minimization problem, and $w_3 \geq 0$

because the third primal constraint has the sign $\geq$. Most importantly, since x_3 is unrestricted in the primal, its corresponding dual constraint is $\boxed{4w_1 - w_3 = 1}$. Hence this example verifies the important duality rule

Unrestricted variable in primal $\iff$ equality constraint in dual.

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7 Practice Problems

Problem 1

Write the dual of

$$\begin{aligned} \text{Maximize} \quad & Z = x_1 + 2x_2, \\ & 2x_1 - 3x_2 \leq 3, \\ & 4x_1 + x_2 \leq -4, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Problem 2

Write the dual of

$$\begin{aligned} \text{Minimize} \quad & Z = 4x_1 + 6x_2 + 18x_3, \\ & x_1 + 3x_3 \geq 3, \\ & x_2 + 2x_3 \geq 5, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Problem 3

Write the dual of

$$\begin{aligned} \text{Maximize} \quad & Z = x_1 + 3x_2, \\ & 3x_1 + 2x_2 \leq 6, \\ & 3x_1 + x_2 = 4, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Problem 4

Write the dual of

$$\begin{aligned}\text{Maximize} \quad & Z = 3x_1 + 5x_2 + 4x_3, \\ & 2x_1 + 3x_2 \leq 8, \\ & 2x_2 + 5x_3 \leq 10, \\ & 3x_1 + 2x_2 + 4x_3 \leq 15, \\ & x_1, x_2, x_3 \geq 0.\end{aligned}$$